Lab 3: Frequency Response of a Common-Emitter BJT Amplifier

Pre-lab

Given. $V_{CC} = 15$ V, $R_1 = 22$ kΩ, $R_2 = 12$ kΩ, $R_C = 3.3$ kΩ, $R_E = 4.3$ kΩ, $R_L = 4.7$ MΩ, $C_1 = C_2 = C_E = 10$ μF. 2N3904 (NPN): $\beta = 150$, $V_A \approx 74$ V, $V_{BE} = 0.7$ V, $V_T = 26$ mV.

DC analysis

At DC the capacitors are open. Replace the divider by its Thevenin equivalent (Figure 1):

$$V_{Th} = 15 \cdot \frac{12}{34} = 5.29 \text{ V}, \quad R_B = 12 \text{ k} \parallel 22 \text{ k} = 7.76 \text{ k}\Omega$$

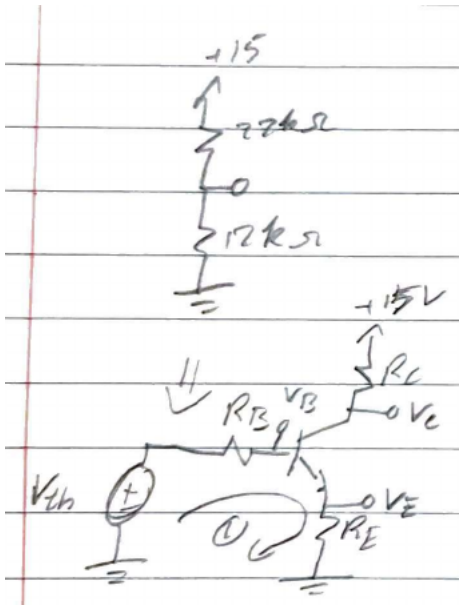


Figure 1: Bias divider and its Thevenin equivalent;①is the KVL loop.

KVL around loop①, then solve for I_B:

$$V_{Th} = I_B R_B + V_{BE} + (\beta + 1) I_B R_E \quad \Rightarrow \quad I_B = \frac{V_{Th} - V_{BE}}{R_B + (\beta + 1) R_E} = 6.99 \text{ μA}$$

$$I_C = \beta I_B = 1.05 \text{ mA}, \quad I_E = (\beta + 1) I_B = 1.06 \text{ mA}$$

$$V_E = I_E R_E = 4.54 \text{ V}, \quad V_B = V_E + V_{BE} = 5.24 \text{ V}, \quad V_C = V_{CC} - I_C R_C = 11.54 \text{ V}$$

$V_{CE} = 7.0$ V and $V_{CB} = 6.3$ V > 0, so the transistor is in forward active. Small-signal parameters:

$$g_m = \frac{I_C}{V_T} = 40.3 \text{ mS}, \quad r_\pi = \frac{\beta}{g_m} = 3.72 \text{ k}\Omega, \quad r_o = \frac{V_A}{I_C} = 70.6 \text{ k}\Omega$$

Task 1: Circuit with parasitic capacitances

C_π is the capacitance between base and emitter and C_μ the capacitance between base and collector (Figure 2).

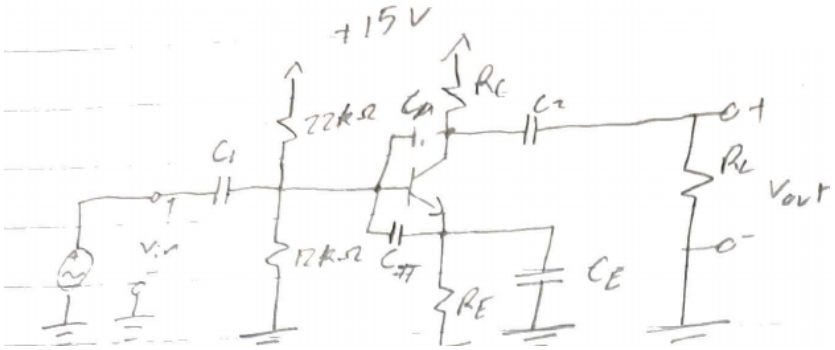


Figure 2: Common-emitter amplifier with C_π (B-E) and C_μ (B-C) added.

Their values follow from the 2N3904 SPICE model parameters: $C_{\text{jc0}} = 3.64$ pF, $\varphi_c = 0.75$ V, $m_c = 0.31$, $\tau_F = 301$ ps and a forward-biased $C_{\text{je}} \approx 6.4$ pF.

$$C_\mu = \frac{C_{\text{jc0}}}{(1 + V_{\text{CB}}/\varphi_c)^{m_c}} = 1.8 \text{ pF}, \quad C_\pi = g_m \tau_F + C_{\text{je}} = 19 \text{ pF}$$

Check: $f_T = g_m/\big(2\pi(C_\pi + C_\mu)\big) = 310$ MHz, close to the 300 MHz in the datasheet.

Task 2: Gain as $f \to 0$

All capacitors are open (Figure 3), so no AC signal reaches the base.

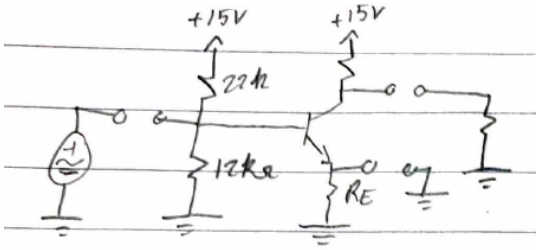


Figure 3: The amplifier as $f \to 0$: every capacitor is open.

$A_v \to 0$ as $f \to 0$. The two zeros at $s = 0$ (from C_1 and C_2) give a +40 dB/dec slope.

Task 3: Gain as $f \to \infty$

All capacitors, including C_π and C_μ, are shorted (Figure 4).

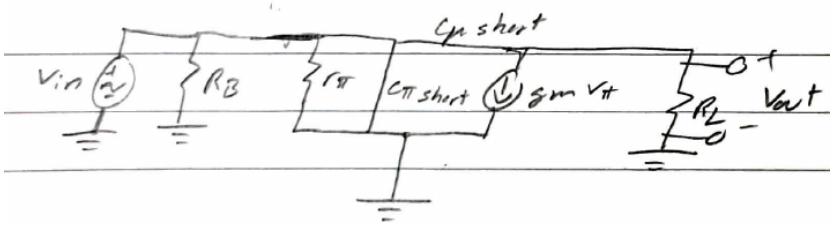


Figure 4: Small-signal circuit as $f \to \infty$.

- The C_π short is in parallel with r_π, so $v_\pi = 0$ and $g_m v_\pi = 0$.
- The C_μ short ties the output node to the base, which the C_π short holds at ground, so $v_{\text{out}} = 0$.

$A_v \to 0$ as $f \to \infty$.

Task 4: Mid-band gain

C_1, C_2, C_E are shorted and only the parasitics C_π, C_μ are open (Figure 5).

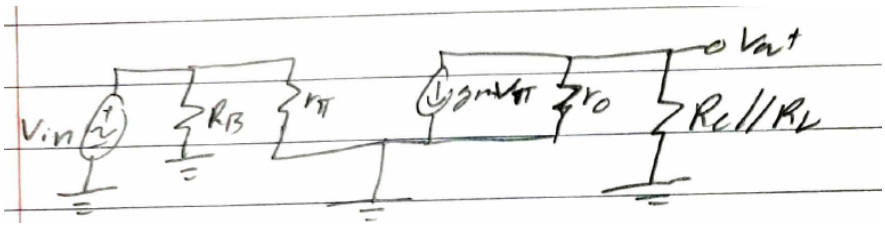


Figure 5: Mid-band small-signal circuit.

$$v_\pi = v_{\text{in}}, \quad r_o \parallel R_C \parallel R_L = 3.15\ \text{k}\Omega, \quad v_{\text{out}} = -g_m v_\pi (R_C \parallel R_L \parallel r_o)$$

$$A_m = \frac{v_{\text{out}}}{v_{\text{in}}} = -g_m (R_C \parallel R_L \parallel r_o) = -(40.3\ \text{mS})(3.15\ \text{k}\Omega) = -127$$

$A_m = -127$, $20 \log_{10} 127 = 42.1$ dB (-133 if r_o is neglected).

Table 1: Mid-band voltage gain (manual Table 2).

Theoretical gain	**Measured gain A_m**
-127 (42.1 dB)	

Tasks 5 to 7: MultiSim

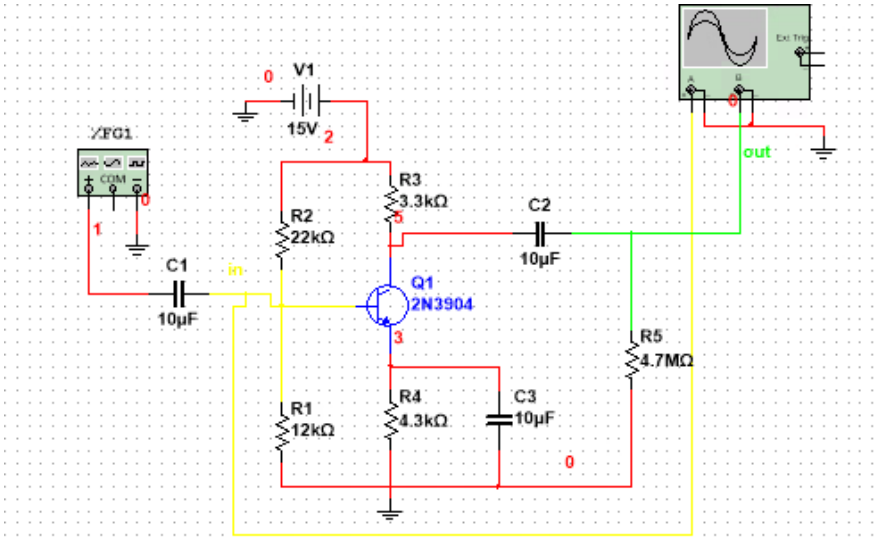


Figure 6: Circuit built in MultiSim (Task 5), with net names shown.

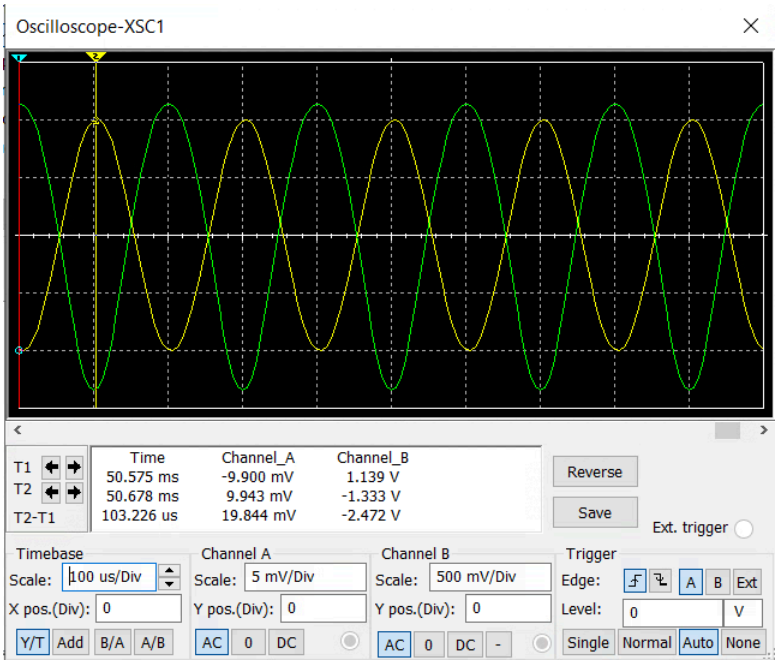
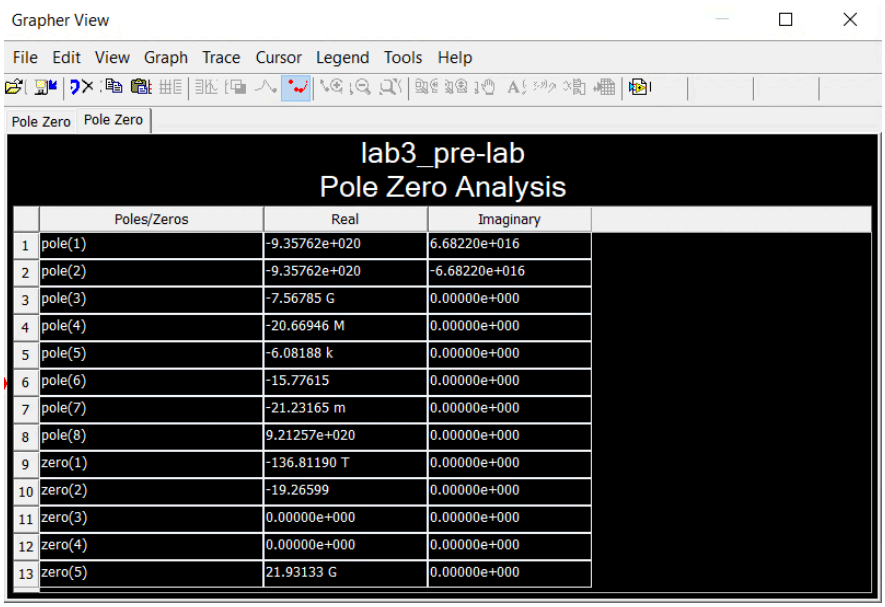


Figure 7: Oscilloscope at 5 kHz: channel A (yellow) is v_{in} at the base, channel B (green) is v_{out}. The cursors give 19.8 mVpp in and 2.47 Vpp out, so $|A_v| \approx 125$, inverted, close to the theoretical -127.

Pole-zero analysis (Task 6): Simulate > Analyses and Simulation > Pole Zero, gain analysis from net 1 (generator side of C_1) to net out, both referenced to ground.



	Poles/Zeros	Real	Imaginary
1	pole(1)	-9.35762e+020	6.68220e+016
2	pole(2)	-9.35762e+020	-6.68220e+016
3	pole(3)	-7.56785 G	0.00000e+000
4	pole(4)	-20.66946 M	0.00000e+000
5	pole(5)	-6.08188 k	0.00000e+000
6	pole(6)	-15.77615	0.00000e+000
7	pole(7)	-21.23165 m	0.00000e+000
8	pole(8)	9.21257e+020	0.00000e+000
9	zero(1)	-136.81190 T	0.00000e+000
10	zero(2)	-19.26599	0.00000e+000
11	zero(3)	0.00000e+000	0.00000e+000
12	zero(4)	0.00000e+000	0.00000e+000
13	zero(5)	21.93133 G	0.00000e+000

Figure 8: MultiSim pole-zero result (s in rad/s).

Table 2: Poles and zeros of $A_v(s) = V_{\text{out}}(s)/V_{\text{in}}(s)$ from MultiSim (Task 7). MultiSim also listed poles near $\pm 9 \times 10^{20}$ rad/s and a zero at -1.4×10^{14} rad/s; these are numerical artifacts and are left out.

	Origin	s **(rad/s)**	$f = \lvert s \rvert / 2\pi$
z_1, z_2	C_1, C_2 block DC	0	0 Hz
z_3	C_E with R_E	-19.3	3.07 Hz
z_4	C_μ feedforward (right half-plane)	$+2.19 \times 10^{10}$	3.49 GHz
p_1	C_2 with R_L	-2.12×10^{-2}	3.38 mHz
p_2	C_1	-15.8	2.51 Hz
p_3	C_E (dominant low-frequency pole)	-6.08×10^{3}	968 Hz
p_4	Miller C_μ and output node (dominant high)	-2.07×10^{7}	3.29 MHz
p_5	Non-dominant high-frequency pole	-7.57×10^{9}	1.20 GHz

Task 8: Bode plot

From Table 2, the straight-line slopes are: +20 dB/dec up to p_2 (2.51 Hz), flat to z_3 (3.07 Hz), +20 dB/dec to p_3 (968 Hz), flat at 42.1 dB to p_4 (3.29 MHz), then −20 dB/dec, −40 dB/dec after p_5 (1.20 GHz) and −20 dB/dec after z_4 (3.49 GHz).

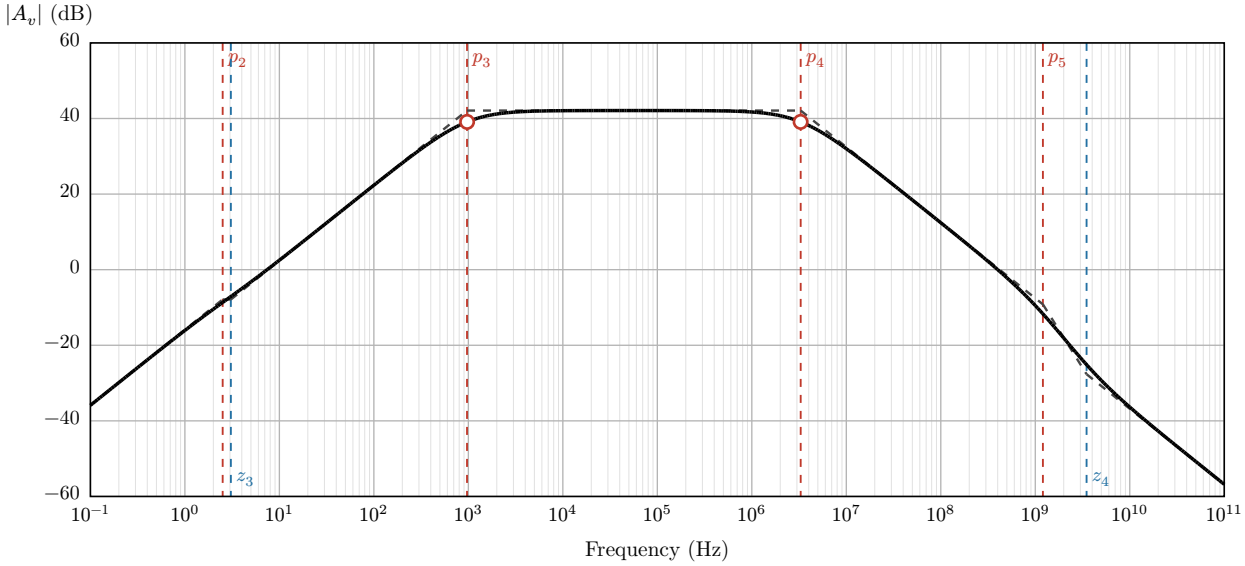


Figure 9: Voltage-gain magnitude from the MultiSim poles and zeros: exact (solid) and straight-line asymptotes (dashed). Red and blue dashed lines mark poles and zeros ($p_1 = 3.4$ mHz is off scale); circles mark the 3-dB points.

$$f_L \approx 970 \text{ Hz}, \quad f_H \approx 3.28 \text{ MHz}$$